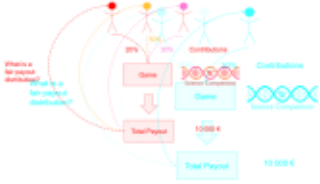
Interpretable Machine Learning

Shapley Values



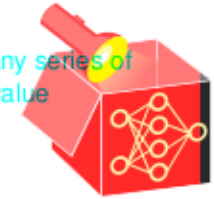
Learning goals

- Learn what game theory is
- Understand the concept behind cooperative games
- Understand the Shapley value in game theory

COOPERATIVE GAMES IN GAME THEORY

Shapley (1951)

- Game theory is the study of strategic games between players, "game" refers to any series of interactions between actors/agents with gains and losses of quantifiable utility value



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COOPERATIVE GAMES IN GAME THEORY

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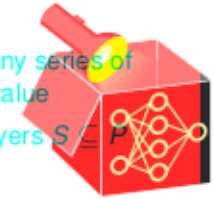

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Interpretable Machine Learning – 1 / 10

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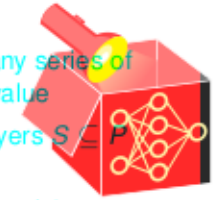

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Interpretable Machine Learning – 1 / 10

COOPERATIVE GAMES IN GAME THEORY

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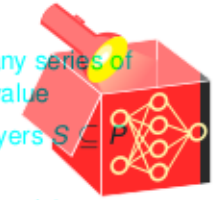

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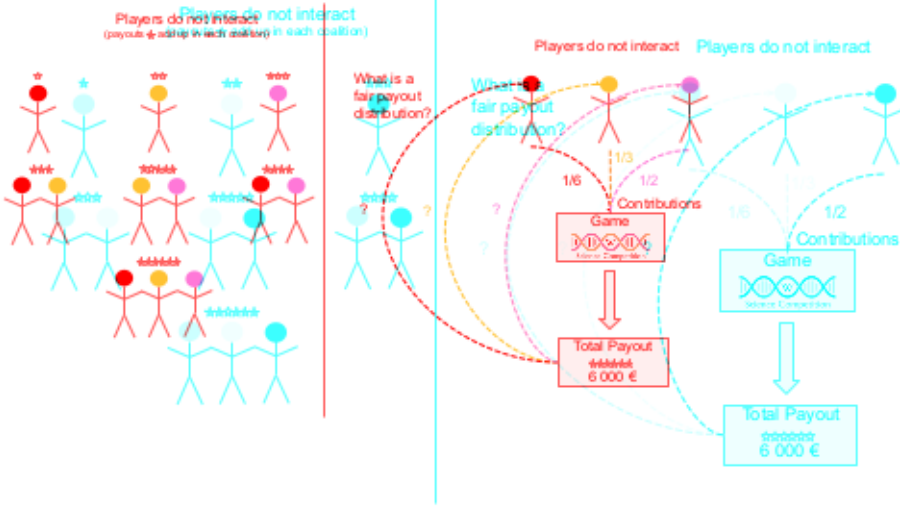
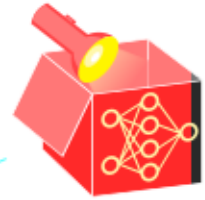
COOPERATIVE GAMES WITHOUT INTERACTIONS





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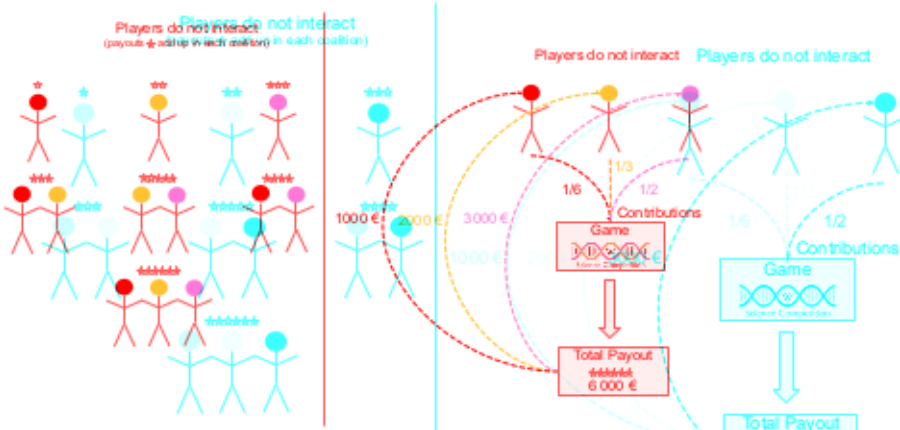
COOPERATIVE GAMES WITHOUT INTERACTIONS



⇒ Fair Payouts are Trivial Without Interactions

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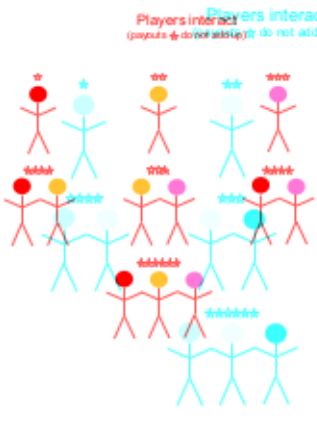
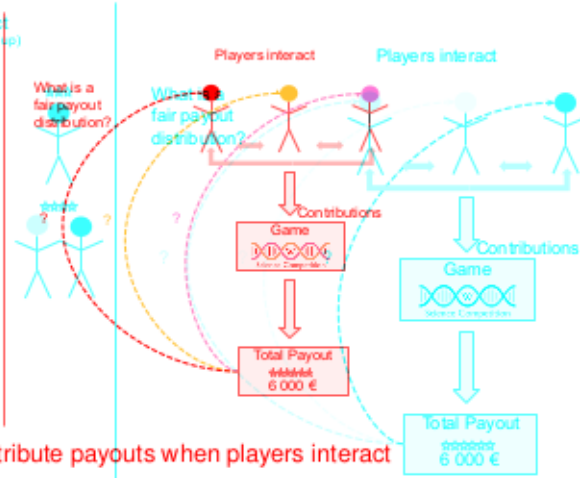
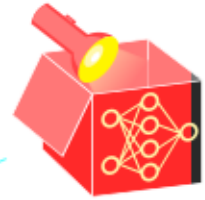
Interpretable Machine Learning – 2 / 10

COOPERATIVE GAMES WITH INTERACTIONS







⇒ Unclear how to fairly distribute payouts when players interact

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Interpretable Machine Learning – 3 / 10

COOPERATIVE GAMES WITH INTERACTIONS

Question: What is a fair payout for player "yellow"?

Idea: Compute marginal contribution of the player of interest across different coalitions

- Compute the total payout of each coalition
- Compute difference in payouts for each coalition with and without player "yellow" (= marginal contribution)
- Average marginal contributions using appropriate weights

Interpretable Machine Learning – 4 / 10

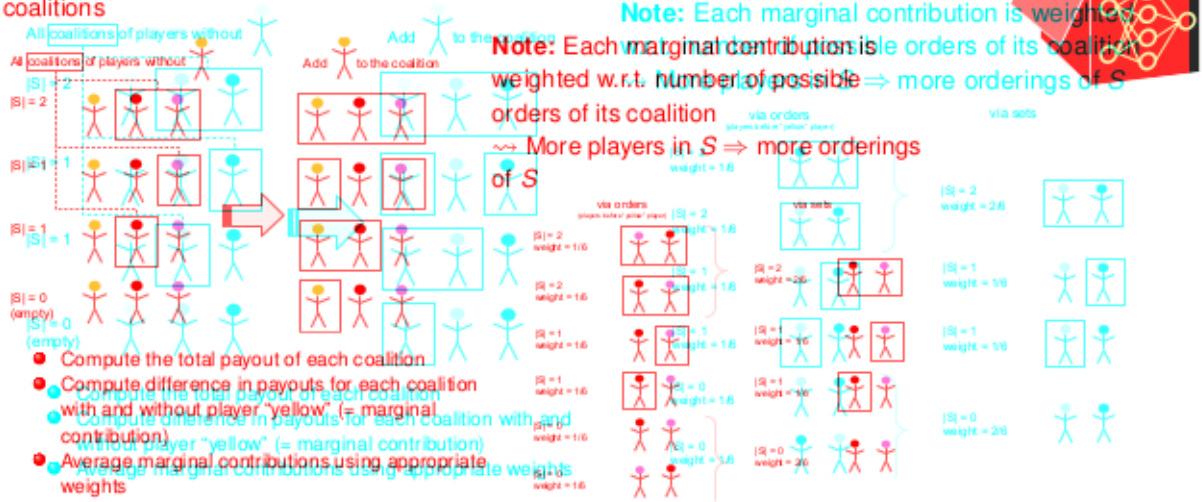
COOPERATIVE GAMES WITH INTERACTIONS

Question: What is a fair payout for player "yellow"?

Idea: Compute marginal contribution of the player of interest across different coalitions

Note: Each marginal contribution is weighted w.r.t. number of possible orders of its coalition

$\rightsquigarrow$ More players in $S \Rightarrow$ more orderings of S



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Interpretable Machine Learning – 4 / 10

SHAPLEY VALUE - SET DEFINITION

This idea refers to the **Shapley value** which assigns a payout value to each player according to its marginal contribution in all possible coalitions.

- Let $v(S \cup \{j\}) - v(S)$ be the marginal contribution of player j to coalition S
  $\rightsquigarrow$ measures how much a player j increases the value of a coalition S

Interpretable Machine Learning – 5 / 10

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Interpretable Machine Learning – 5 / 10

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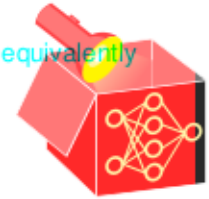
- Shapley value via **set definition** (weighting via multinomial coefficient):

$$\phi_j = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!(|P| - |S| - 1)!}{|P|!} (v(S \cup \{j\}) - v(S))$$

Interpretable Machine Learning – 5 / 10

SHAPLEY VALUE - ORDER DEFINITION



The Shapley value was introduced as summation over sets $S \subseteq P \setminus \{j\}$, but it can be equivalently defined as a summation of all orders of players:

$$\phi_j = \frac{1}{|P|!} \sum_{\tau \in \Pi} (v(S_j^\tau \cup \{j\}) - v(S_j^\tau))$$

- Π: All possible orders of players (we have $|P|!$ in total)

Interpretable Machine Learning – 6 / 10

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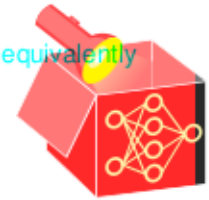
Interpretable Machine Learning – 6 / 10

SHAPLEY VALUE - ORDER DEFINITION



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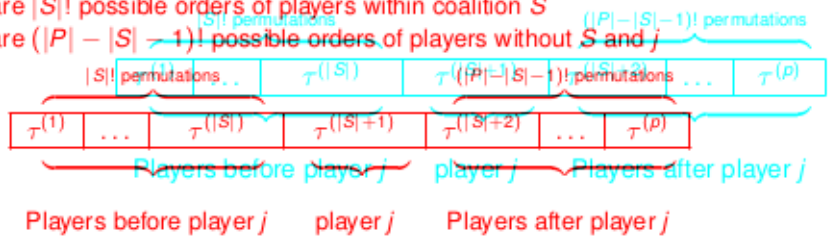
    - $\rightsquigarrow$ For order $\tau = (3, 1, 2)$ and player of interest $j = 3 \Rightarrow S_j^\tau = \emptyset$
    - $\rightsquigarrow$ In set definition, it has the weight $\frac{(|S|-1)!(|P|-|S|)!}{|P|!} = \frac{2!0!}{3!} = \frac{2}{6}$
- Order definition: Marginal contribution of orders that yield set $S = \{1, 2\}$ is summed twice
  - $\rightsquigarrow$ In set definition, it has the weight $\frac{2!(3-2-1)!}{3!} = \frac{2 \cdot 0!}{6} = \frac{2}{6}$

SHAPLEY VALUE - COMMENTS ON ORDER DEFINITION

- Order and set definition are equivalent
- Reason: The number of orders which yield the same coalition S is $|S|!(|P| - |S| - 1)!$
  - $\Rightarrow$ There are $|S|!$ possible orders of players within coalition S
  - $\Rightarrow$ There are $(|P| - |S| - 1)!$ possible orders of players without S and j



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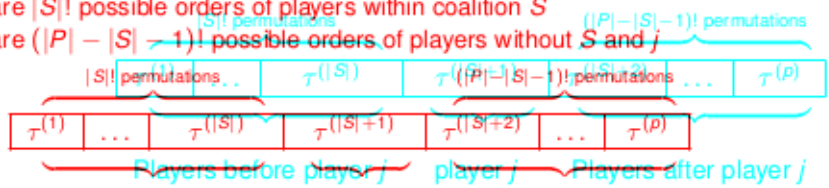

- Relevance of the order definition: Approximate Shapley values by sampling permutations
  ⇝ randomly sample a fixed number of M permutations and average them:

$$\phi_j = \frac{1}{M} \sum_{\tau \in \Pi_M} \left(v(S_j^\tau \cup \{j\}) - v(S_j^\tau) \right)$$

where $\Pi_M \subset \Pi$ is a random subset of Π containing only M orders of players

WEIGHTS FOR MARGINAL CONTRIBUTION - ILLUSTRATION

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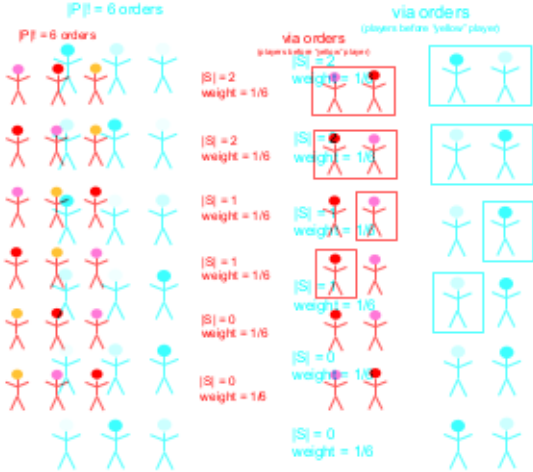

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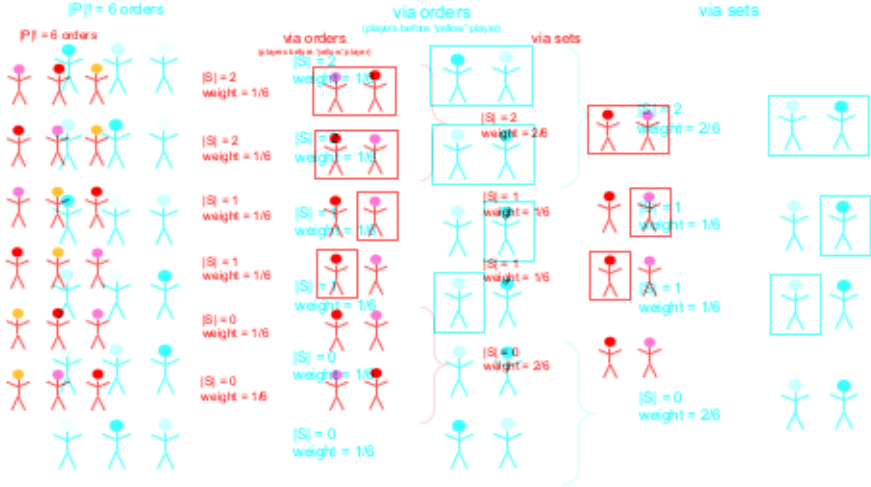

Interpretable Machine Learning – 8 / 10

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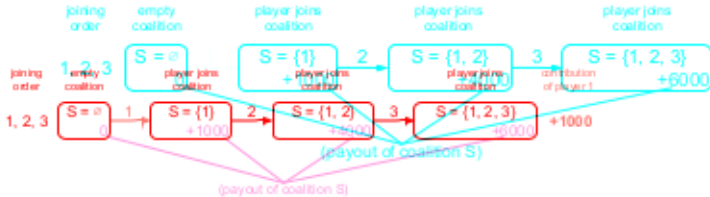

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SHAPLEY VALUES - ILLUSTRATION

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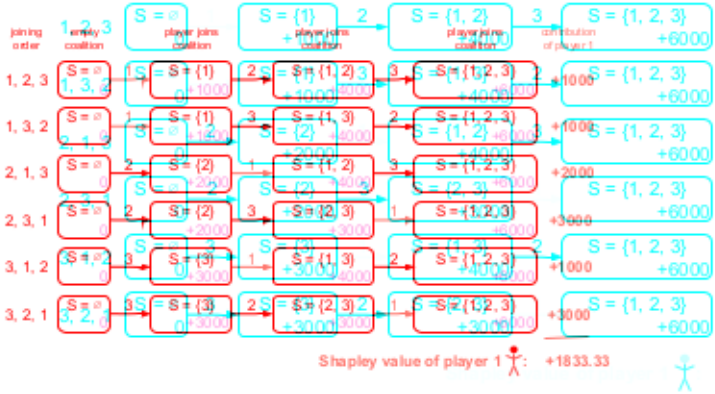
- Produce all possible joining orders of player coalitions
- Measure and average the difference in payout after player 1 enters the coalition



Interpretable Machine Learning – 9 / 10

SHAPLEY VALUES - ILLUSTRATION

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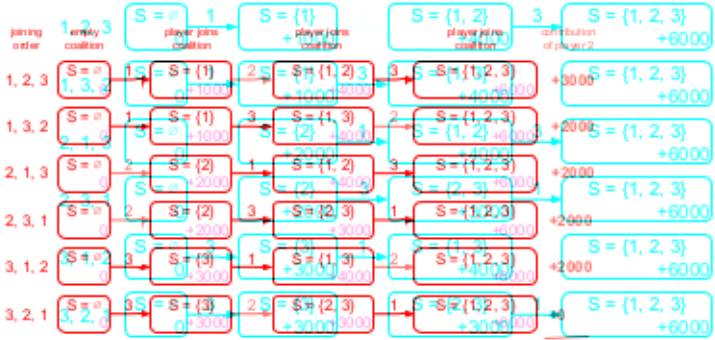

Interpretable Machine Learning – 9 / 10

SHAPLEY VALUES - ILLUSTRATION

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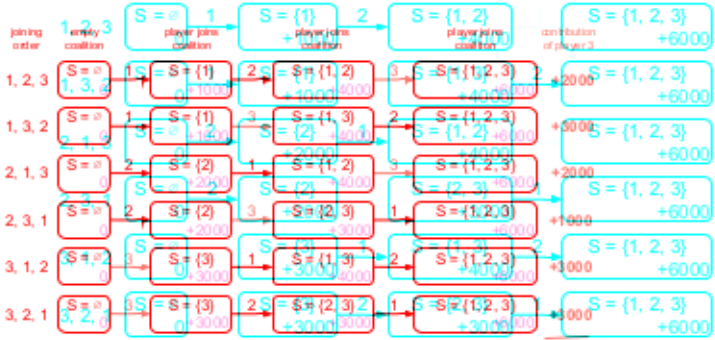
- Produce all possible joining orders of player coalitions
- Measure and average the difference in payout after player 3 enters the coalition



Interpretable Machine Learning – 9 / 10

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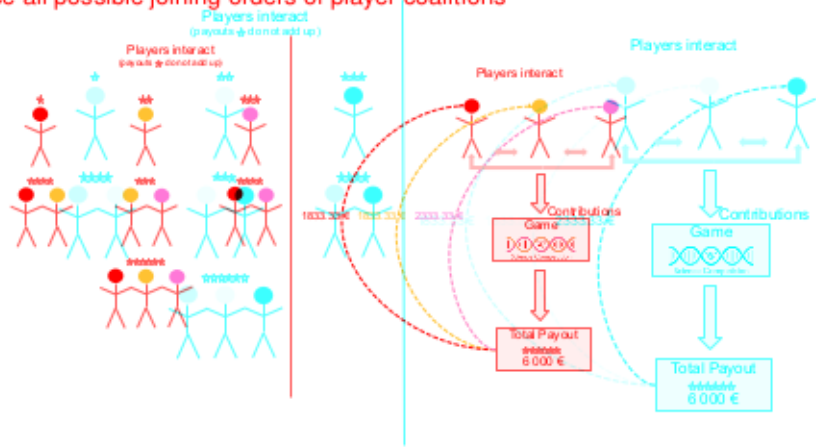

Interpretable Machine Learning – 9 / 10

AXIOMS OF FAIR PAYOUTS

Why is this a fair payout solution?

One possibility to define fair payouts are the following axioms for a given value function v:

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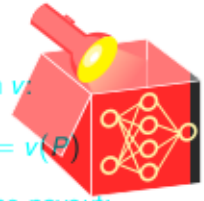
- **Dummy/Null Player**: Payout is 0 for players who don't contribute to the value of any coalition:
  If $v(S \cup \{j\}) = v(S) \quad \forall \quad S \subseteq P \setminus \{j\}$, then $\phi_j = 0$



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AXIOMS OF FAIR PAYOUTS

Why is this a fair payout solution?

One possibility to define fair payouts are the following axioms for a given value function v:

- **Efficiency**: Player contributions add up to the total payout of the game: $\sum_{j=1}^{p} \phi_j = v(P)$
- **Symmetry**: Players $j, k \in P$ who contribute the same to any coalition get the same payout:
  If $v(S \cup \{j\}) = v(S \cup \{k\})$ for all $S \subseteq P \setminus \{j, k\}$, then $\phi_j = \phi_k$
- **Dummy/Null Player**: Payout is 0 for players who don't contribute to the value of any coalition:
  If $v(S \cup \{j\}) = v(S) \quad \forall \quad S \subseteq P \setminus \{j\}$, then $\phi_j = 0$
- **Additivity**: For a game v with combined payouts $v(S) = v_1(S) + v_2(S)$, the payout is the sum of payouts: $\phi_{j,v} = \phi_{j,v_1} + \phi_{j,v_2}$

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